SSC MTS 2024 question paper with answers

13 November 2024 · Shift 1

80 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 10 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 14 questions

Q.1

If the number of working staff in the company is increased by 20% and the salary per head is decreased by 20%, then the percentage of change in the total salary is:

(A) decrease of 20%
(B) increase of 20%
(C) decrease of 4%
(D) increase of 4%

Q.2

Pankaj bought a watch at a price range from ₹1,500 to ₹3,000 which are sold at prices ranging from ₹2,500 to ₹3,500. If Pankaj sells such 10 watches, then the greatest possible profit (in ₹) he made with the transaction is:

(A) 19,000
(B) 23,854
(C) 21,765
(D) 20,000

Q.4

A 250-metre-long train running at a speed of 100 km/h crosses another train coming from the opposite direction at a speed of 62 km/h in 10 seconds. What is the length of the second train?

(A) 230 metres
(B) 200 metres
(C) 240 metres
(D) 270 metres

Q.5

A sum of ₹12,000 becomes double at simple interest in 7 years at a certain rate of simple interest. Find the time (in years) it takes for the amount to become ₹60,000 at the same rate of interest.

(A) 28
(B) 30
(C) 20
(D) 25

Q.6

A spherical metal ball is melted into a shape of cylinder mounted by a cone such that the base of cone exactly overlaps the top base of cylinder. The diameter of the cylinder and sphere is 4.2 m each. What is the height (in m) of the cone if the height of cylinder is 1.4 m?

(A) 0.7
(B) 1.4
(C) 4.2
(D) 2.1

Q.11

The average of 7 numbers is 55. If the first 4 numbers are 48, 62, 135 and 74, then what will be the average of the remaining numbers?

(A) 22
(B) 31
(C) 25
(D) 19

Q.13
Two trains of lengths 310 m and 340 m, respectively, are 285 m apart. They start moving towards each other on parallel tracks, at the speeds of 151 km/h and 155 km/h, respectively. In how much time will the trains cross each other completely?

(A) 13 s
(B) 10 s
(C) 11 s
(D) 12 s

Q.14
Find the number that must be added to the terms of the ratio 11 : 29 to make it equal to 11 : 20.

(A) 12
(B) 10
(C) 11
(D) 9

Q.15
A right circular solid metallic cone is 7.2 cm high and radius of its base 3.5 cm. It is melted and recast into two right circular solid metallic cones equal in all respects, each with base radius 2.1 cm and height 'h'. What is the value of 'h' (in cm)?

(A) 14
(B) 10
(C) 21
(D) 12

Q.16
A shopkeeper of a subsidised canteen mixed three varieties of rice, X, Y and Z, in the ratio of 4 : 1 : 3 to make a packet of 4 kg of rice. The cost prices per kg of X, Y and Z varieties of rice are ₹2, ₹4 and ₹6, respectively. What is the cost price (in ₹) of the packet of rice?

(A) 10
(B) 15
(C) 30
(D) 18

Q.17
Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 7% per year rate of interest for 2 years.

(A) 390
(B) 290
(C) 310
(D) 350

Q.18
The average weight of 6 children increases by 1.5 kg when a new child comes in place of one of them weighing 12 kg. The weight of the new child is:

(A) 21 kg
(B) 14 kg
(C) 22 kg
(D) 14.5 kg

Q.19
The average height of a group of 10 students is 172 cm. Two more students of height 178 cm and 165 cm joined the group and 3 students of average height 160 cm leave the group. What is the new average height (in cm, rounded off to 1 decimal place) of the group after the changes?

(A) 174.6
(B) 170.5
(C) 175.9
(D) 169.7

Q.20
Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 7% per year rate of interest for 4 years.

(A) 640
(B) 660
(C) 700
(D) 740

Reasoning Ability and Problem Solving 16 questions

Q.1

Which of the following equations is correct by interchanging the '+' and '×' signs and the '4' and '5' numbers?

(A) 20 + 5 × 4 = 85
(B) 5 + 15 × 4 = 50
(C) 5 × 4 + 10 = 40
(D) 10 × 4 + 5 = 60

Q.2

What will come in place of the question mark (?) in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 392 × 7 – 2 ÷ 12 + 80 = ?

(A) 0
(B) 2
(C) 8
(D) 4

Q.3

In a certain code language, 'DAYS' is coded as '3917' and 'YOLK' is coded as '2496'.What is the code for 'Y' in that language?

(A) 1
(B) 4
(C) 2
(D) 9

Q.5

4 is related to 70 following a certain logic. Following the same logic, 8 is related to 518. To which ofthe following is 6 related, following the same logic?(NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1and 3 is not allowed)

(A) 226
(B) 219
(C) 222
(D) 216

Q.6

In a certain code language, 'YATCH' is coded as 'RTMVA' and 'CAT' is coded as 'VTM'. What is the code for 'ANAMOLIES' in the given code language?

(A) TGTEHEBXL
(B) TGTFGEBXL
(C) TGTFHEBXL
(D) TGSFHEBXL

Q.8

What should come in place of the question mark (?) in the given series? 118 122 130 142 158 ?

(A) 178
(B) 182
(C) 186
(D) 166

Q.11

LO15 is related to OR30 in a certain way. In the same way, SV25 is related to VY50. To which of the following is TW35 related, following the same logic?

(A) WZ70
(B) WA70
(C) WZ17
(D) AZ17

Q.12

This question has two statements followed by two conclusions numbered I and II. You have to take the given statements to be true even if they seem to be at variance from commonly known facts. You have to decide which of the given conclusions logically follow(s) from the given statements. Statements: All arrows are swords. Some swords are dragons. Conclusions (I) Some swords are arrows. (II) Some dragons are not swords.

(A) Neither conclusion (I) nor (II) follows
(B) Both conclusions (I) and (II) follow
(C) Only conclusion (II) follows
(D) Only conclusion (I) follows

Q.13

Seven people A, B, C, D, E, F and G are sitting around a circular table, facing the centre (but not necessarily in the same order). Exactly three people are sitting between E and B when counted from the right of E. Both G and D are immediate neighbours of F. C is sitting to the immediate left of A. D and E are not sitting next to each other. The position of how many persons will remain unchanged, excluding A, if all the persons are arranged in the English alphabetical order, in clockwise direction, starting from A?

(A) 1
(B) 2
(C) 4
(D) 3

Q.14

Three statements are given, followed by two conclusions numbered 1, and 2. Assuming the statements to be true, even if they seem to be at variance with commonly known facts, decide which of the conclusions logically follow(s) from the statements. Statements: No Doctor is a Nurse; Some Nurses are Patients; Some Patients are Medicines Conclusions: 1. Some Patients are not Doctors 2. Some Nurses are Medicines

(A) Neither conclusion 1 nor 2 follows
(B) Only conclusion 1 follows
(C) Only conclusion 2 follows
(D) Both conclusions 1 and 2 follow

Q.15

A, B, C, D, E, F and G are sitting around a circular table facing the centre (but not necessarily in the same order). G sits second to the left of F, who is third to the left of D. C sits third to the right of B, who is not an immediate neighbour of E. A sits second to the right of B. Who is immediate right of E?

(A) A
(B) D
(C) G
(D) B

Q.16

NLRP is related to USYW in a certain way based on the English alphabetical order. In the same way, KIOM is related to RPVT. To which of the following is GEKI related, following the same logic?

(A) NLPR
(B) LNRP
(C) NLRP
(D) LNPR

Q.17

What should come in place of the question mark (?) in the given series based on the English alphabetical order? RJK, PMI, NPG, LSE, ?

(A) KVD
(B) KWC
(C) JVC
(D) KVC

Q.18

In a certain code language, 'MINT' is coded as '9735' and 'NEAR' is coded as '4568'. What is the code for 'N' in the given code language?

(A) 3
(B) 5
(C) 6
(D) 4

Q.19

In a certain code language, ‘FISH’ is coded as ‘6482’ and ‘SANE’ is coded as ‘7691’. What is the code for ‘S’ in the given code language?

(A) 8
(B) 7
(C) 6
(D) 9

Q.20

19 is related to 39 following a certain logic. Following the same logic, 1 is related to 3. To which of the following is 3 related following the same logic?(NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1and 3 is not allowed)

(A) 7
(B) 9
(C) 11
(D) 8

General Awareness 25 questions

Q.1

Which is the jet stream that brings Western disturbances in India?

(A) Easterly jet stream
(B) Westerly jet stream
(C) South polar jet stream
(D) North polar jet stream

Q.2

Aparna Satheesan, a recipient of 2021 Abhinandan Saroja National award by National Institute of Classical Dance, is an exponent of _____ and Kuchipudi.

(A) Bharatanatyam
(B) Kathak
(C) Manipuri
(D) Mohiniyattam

Q.3

According to Census of India 2011, which pair of states recorded the highest population density (persons/km^2) in India?

(A) Bihar and Kerala
(B) West Bengal and Uttar Pradesh
(C) Bihar and West Bengal
(D) Uttar Pradesh and Haryana

Q.4

Identify the place where the West Central Railway Zone headquarters is located.

(A) Jabalpur
(B) Jodhpur
(C) Bhopal
(D) Jaipur

Q.5

Which of the following Rashtrakuta rulers performed the ceremony called ‘Hiranya Garbha’?

(A) Krishna I
(B) Dantidurga
(C) Govind III
(D) Dhruva I

Q.6

Who can exercise a casting vote in the case of an equality of votes in Lok Sabha?

(A) Speaker of Lok Sabha
(B) President
(C) Prime Minister
(D) Vice President

Q.7

The autobiography of Sam Pitroda is called:

(A) In the afternoon of time
(B) Dreaming Big: My Journey to Connect India
(C) The Substance and The Shadow
(D) Beyond the lines

Q.8

On which of the following dates did the British Empire transfer its powers to the Constituent Assembly in India?

(A) 2nd October
(B) 15th August
(C) 14th November
(D) 26th January

Q.9
The Right to Freedom of Religion is covered under which Articles of the Constitution of India?

(A) Articles 25-28
(B) Articles 21-24
(C) Articles 29-32
(D) Articles 24-27

Q.10
A little child is a very picky eater and hates drinking milk and milk products. When he lost his baby tooth, permanent teeth took a very long time to come up in its place. What could be the probable reason behind this?

(A) Vitamin Deficiency
(B) Sugar Deficiency
(C) Calcium Deficiency
(D) Protein Deficiency

Q.11
The 'Quit India Movement' was launched under the leadership of _____.

(A) Jawaharlal Nehru
(B) Mahatma Gandhi
(C) Subhash Chandra Bose
(D) Muhammad Ali Jinnah

Q.12
On May 23, 2023, who assumed the position of Chairperson at the Competition Commission of India?

(A) Subhash Chandra Garg
(B) Neha Bhatia
(C) NK Singh
(D) Ravneet Kaur

Q.13
Which of the following Indian philosophical systems and its widely acknowledged proponents is INCORRECTLY paired?

(A) Yoga - Patanjali
(B) Purva Mimansa - Jaimini
(C) Nyaya - Gotama
(D) Vaisheshika - Vyasa

Q.14
Which of the following statements is/are true about public goods? I. Rivalrous in consumption II. Non-excludability III. Positive marginal cost

(A) Only I
(B) Only II and III
(C) Only I and III
(D) Only II

Q.15
Which of the following folk dances of Jharkhand, Odisha and West Bengal region is on the Representative List of the Intangible Cultural Heritage of Humanity?

(A) Kummi
(B) Chhau
(C) Karma Naach
(D) Dadra

Q.16
Where did the IBA Women's World Boxing Championships 2023 take place in India?

(A) Mumbai
(B) New Delhi
(C) Kolkata
(D) Bengaluru

Q.17
The rifle (in the shooting sport) is single-loaded in _____-millimetre calibre (the inner diameter of a gun barrel), which is used across all events.

(A) 5.6
(B) 4.8
(C) 4.5
(D) 5.1

Q.18

In Harappan civilisation, which material was NOT used in the construction of buildings?

(A) Wood
(B) Burnt mud-bricks
(C) Iron
(D) Gypsum mortar

Q.19

The National Food Security Act (NFSA) was enacted in which year by the Government of India?

(A) 2001
(B) 2013
(C) 2008
(D) 2005

Q.20

As of July 2023, how many products were notified under the 'One District, One Product' scheme as per the information provided by the Ministry of Food Processing Industries, Government of India?

(A) 132
(B) 125
(C) 135
(D) 137

Q.21

Vibration or wave caused by the sudden movement of tectonic plates within the earth that can travel through and around the earth is called:

(A) Seismic Wave
(B) Main Shock
(C) Foreshock
(D) Upheaval

Q.22

Which of the following agreements led Mahatma Gandhi to attend the Second Round Table Conference?

(A) Gandhi Irwin Pact
(B) Poona Pact
(C) Lucknow Pact
(D) Pact of Paris

Q.23

What is the primary source of energy for most ecosystems on Earth?

(A) Geothermal heat
(B) Tidal energy
(C) Wind
(D) Sunlight

Q.24

Who was the progenitor of the prolific Jaipur-Atrauli gharana?

(A) Ustad Alladiya Khan
(B) Ustad Vilayat Khan
(C) Ustad Ali Akbar Khan
(D) Ustad Mustaq Ali Khan

Q.25

0.25 g of an organic compound gave 0.20 g of CO_2 after complete combustion. The percentage composition of carbon is _____.

(A) 4.60%
(B) 21.82%
(C) 80.00%
(D) 78.18%

English Language and Comprehension 25 questions

Q.1

Select the most appropriate option to fill in the blank. The philanthropist's _____ donations significantly improved the quality of life in the community.

(A) substantial
(B) meagre
(C) parsimonious
(D) redundant

Q.2

Select the INCORRECTLY spelt word..

(A) Peaceful
(B) Lovable
(C) Joyful
(D) Faithfull

Q.3

Select the most appropriate option to fill in the blank. She is _____ of spiders.

(A) interest
(B) afraid
(C) fear
(D) frightening

Q.4

Select the most appropriate meaning of the given idiom. Dig up dirt

(A) To uncover negative information about someone or something.
(B) Find and disclose evidence
(C) Enquire about a missing person
(D) Hide and protect vital evidence

Q.5

Select the most appropriate option to fill in the blank. Following weeks of intense training, the athlete _____ in the championship.

(A) were competing
(B) have been competing
(C) competed
(D) have competed

Q.6

Select the INCORRECTLY spelt word.

(A) Obedence
(B) Aggravate
(C) Fabulous
(D) Tremendous

Q.7

Select the grammatically correct sentence from the given options. These are the works of a great poet Tagore.

(A) These are the works of that great poet Tagore.
(B) These are the works of an great poet Tagore.
(C) These are the works of this great poet Tagore.
(D) These are the works of the great poet Tagore.

Q.8

Select the option that is OPPOSITE in meaning to the underlined word in the given sentence. The state played only a “modest” part in the outstanding advancement of modern morality.

(A) Unimpressive
(B) Boastful
(C) Moderate
(D) Humble

Q.9

Select the correct spelling of the incorrectly spelt word in the given sentence. She is an indepandent writer.

(A) Independent
(B) Indipendant
(C) Indipendent
(D) Independant

Q.10

Select the option that best expresses the meaning of the given idiom. When pigs fly

(A) In unity
(B) In due course of time
(C) Just impossible
(D) Correct approach

Q.11

Select the most appropriate homophone to fill in the blank.While learning to drive, Shivani met with an accident because she couldn't hit the _____.

(A) break
(B) brake
(C) brick
(D) brack

Q.12

Select the option that can be used as a one-word substitute for the underlined word. "He was notorious for his “avaricious” behaviour, always seeking ways to accumulate more wealth."

(A) notorious
(B) greedy
(C) sloth
(D) generous

Q.13

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. The residents of Rameswaram / about the 1940s were quite / orthodox, but were also / tolerant of other communities.

(A) about the 1940s were quite
(B) orthodox, but were also
(C) The residents of Rameswaram
(D) tolerant of other communities

Q.14

Select the most appropriate option to fill in the blank. We are planning to see an _____ perform a symphony.

(A) opera
(B) orchestra
(C) actor
(D) actress

Q.15

Select the most appropriate option to fill in the blank. The cat likes to _____ in the sun during the afternoon.

(A) lye
(B) lei
(C) lie
(D) lay

Q.16

Select the most appropriate option to fill in the blank. The movie was so boring that I fell _____.

(A) slept
(B) sleep
(C) asleep
(D) sleeping

Q.17

The following sentence has been divided into parts. One of them contains an error. Select the part that contains the error from the given options. After months of / meticulous planning and hard work, / a team successfully launched / their innovative product to the market.

(A) their innovative product to the market.
(B) a team successfully launched
(C) meticulous planning and hard work,
(D) After months of

Q.18

Parts of the following sentence have been given as options. Select the option that contains an error. Dr. Oppenheimer is the exceptional movie of all the movies in recent times.

(A) of all the movies
(B) Dr. Oppenheimer is
(C) in recent times
(D) the exceptional movie

Q.19

Select the most appropriate ANTONYM of the given word. Cacophony

(A) Discord
(B) Noise
(C) Dissonance
(D) Harmony

Q.20

Select the grammatically correct sentence from the given options.

(A) A salesman insisted that the payment against an purchase should be done by using the bank card.
(B) The salesman insisted that the payment against the purchase should be done by using a bank card.
(C) The salesman insisted that a payment against the purchase should be done by using an bank card.
(D) A salesman insisted that a payment against a purchase should be done by using a bank card.

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. In terms of technical difficulty, Nanga Parbat, the world's ninth (1)_____ peak, competes with K2. The first (2)_____ to the summit follows a narrow ridge. The 15,000-foot Rupal Face, a single wall of rock and ice, is located on the mountain's southern side. Climbers have dubbed the Nanga Parbat, which (3)_____ as ‘naked mountain’, ‘Killer Mountain’. This gives Annapurna a (4)_____ of 32%, making it the most statistically dangerous of the world’s 8,000-metre peaks. It is an avalanche-prone area with some technical difficulty. Climbers’ misery is (5)_____ by insufficient local support, in addition to unpredictable weather and inaccessibility.

Select the most appropriate option to fill in blank number 1.

(A) deepest
(B) lengthiest
(C) highest
(D) fattest

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. In terms of technical difficulty, Nanga Parbat, the world's ninth (1)_____ peak, competes with K2. The first (2)_____ to the summit follows a narrow ridge. The 15,000-foot Rupal Face, a single wall of rock and ice, is located on the mountain's southern side. Climbers have dubbed the Nanga Parbat, which (3)_____ as ‘naked mountain’, ‘Killer Mountain’. This gives Annapurna a (4)_____ of 32%, making it the most statistically dangerous of the world’s 8,000-metre peaks. It is an avalanche-prone area with some technical difficulty. Climbers’ misery is (5)_____ by insufficient local support, in addition to unpredictable weather and inaccessibility.

Select the most appropriate option to fill in blank number 2.

(A) rise
(B) ascent
(C) lift
(D) launch

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. In terms of technical difficulty, Nanga Parbat, the world's ninth (1)_____ peak, competes with K2. The first (2)_____ to the summit follows a narrow ridge. The 15,000-foot Rupal Face, a single wall of rock and ice, is located on the mountain's southern side. Climbers have dubbed the Nanga Parbat, which (3)_____ as ‘naked mountain’, ‘Killer Mountain’. This gives Annapurna a (4)_____ of 32%, making it the most statistically dangerous of the world’s 8,000-metre peaks. It is an avalanche-prone area with some technical difficulty. Climbers’ misery is (5)_____ by insufficient local support, in addition to unpredictable weather and inaccessibility.

Select the most appropriate option to fill in blank number 3.

(A) transliterates
(B) transposed
(C) translates
(D) transcribes

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. In terms of technical difficulty, Nanga Parbat, the world's ninth (1)_____ peak, competes with K2. The first (2)_____ to the summit follows a narrow ridge. The 15,000-foot Rupal Face, a single wall of rock and ice, is located on the mountain's southern side. Climbers have dubbed the Nanga Parbat, which (3)_____ as ‘naked mountain’, ‘Killer Mountain’. This gives Annapurna a (4)_____ of 32%, making it the most statistically dangerous of the world’s 8,000-metre peaks. It is an avalanche-prone area with some technical difficulty. Climbers’ misery is (5)_____ by insufficient local support, in addition to unpredictable weather and inaccessibility.

Select the most appropriate option to fill in blank number 4.

(A) economic rate
(B) frailty rate
(C) fatality rate
(D) mortal rate

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. In terms of technical difficulty, Nanga Parbat, the world's ninth (1)_____ peak, competes with K2. The first (2)_____ to the summit follows a narrow ridge. The 15,000-foot Rupal Face, a single wall of rock and ice, is located on the mountain's southern side. Climbers have dubbed the Nanga Parbat, which (3)_____ as ‘naked mountain’, ‘Killer Mountain’. This gives Annapurna a (4)_____ of 32%, making it the most statistically dangerous of the world’s 8,000-metre peaks. It is an avalanche-prone area with some technical difficulty. Climbers’ misery is (5)_____ by insufficient local support, in addition to unpredictable weather and inaccessibility.

Select the most appropriate option to fill in blank number 5.

(A) perfectly
(B) increasingly
(C) exacerbated
(D) worst

Answer key

Numerical and Mathematical Ability

Q.1 **(C)**	Q.2 **(D)**	Q.4 **(B)**	Q.5 **(A)**	Q.6 **(C)**	Q.11 **(A)**
Q.13 **(C)**	Q.14 **(C)**	Q.15 **(B)**	Q.16 **(B)**	Q.17 **(D)**	Q.18 **(A)**
Q.19 **(C)**	Q.20 **(C)**				

Reasoning Ability and Problem Solving

Q.1 **(A)**	Q.2 **(A)**	Q.3 **(D)**	Q.5 **(C)**	Q.6 **(C)**	Q.8 **(A)**
Q.11 **(A)**	Q.12 **(D)**	Q.13 **(A)**	Q.14 **(B)**	Q.15 **(C)**	Q.16 **(C)**
Q.17 **(C)**	Q.18 **(B)**	Q.19 **(C)**	Q.20 **(A)**		

General Awareness

Q.1 **(B)**	Q.2 **(A)**	Q.3 **(C)**	Q.4 **(A)**	Q.5 **(B)**	Q.6 **(A)**
Q.7 **(B)**	Q.8 **(B)**	Q.9 **(A)**	Q.10 **(C)**	Q.11 **(B)**	Q.12 **(D)**
Q.13 **(D)**	Q.14 **(D)**	Q.15 **(B)**	Q.16 **(B)**	Q.17 **(A)**	Q.18 **(C)**
Q.19 **(B)**	Q.20 **(D)**	Q.21 **(A)**	Q.22 **(A)**	Q.23 **(D)**	Q.24 **(A)**
Q.25 **(B)**					

English Language and Comprehension

Q.1 **(A)**	Q.2 **(D)**	Q.3 **(B)**	Q.4 **(A)**	Q.5 **(C)**	Q.6 **(A)**
Q.7 **(D)**	Q.8 **(B)**	Q.9 **(A)**	Q.10 **(C)**	Q.11 **(B)**	Q.12 **(B)**
Q.13 **(A)**	Q.14 **(B)**	Q.15 **(C)**	Q.16 **(C)**	Q.17 **(B)**	Q.18 **(D)**
Q.19 **(D)**	Q.20 **(B)**	Q.21 **(C)**	Q.22 **(B)**	Q.23 **(C)**	Q.24 **(C)**
Q.25 **(C)**					

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (C) decrease of 4%

Total salary = staff × salary per head, so it becomes 1.2 × 0.8 = 0.96 of what it was: a decrease of 4%.

Q.2 · Answer (D) 20,000

The profit is greatest when each watch is bought at the lowest price and sold at the highest: (3,500 – 1,500) × 10 = ₹20,000.

Q.4 · Answer (B) 200 metres

Relative speed = 100 + 62 = 162 km/h = 45 m/s, so in 10 seconds the trains cover 450 m between them. The second train is 450 – 250 = 200 metres long.

Q.5 · Answer (A) 28

The interest equals the sum in 7 years. To become ₹60,000 the sum must earn 4 times itself, ₹48,000, which takes 4 × 7 = 28 years.

Q.6 · Answer (C) 4.2

The volumes are equal, with radius 2.1 m throughout: $(4/3)\pi \times 2.1^3 = \pi \times 2.1^2 \times 1.4 + (1/3)\pi \times 2.1^2 \times h$. Dividing by $\pi \times 2.1^2$: 2.8 = 1.4 + h/3, so h = 4.2 m.

Q.11 · Answer (A) 22

The 7 numbers total 7 × 55 = 385 and the first four 48 + 62 + 135 + 74 = 319, so the other three total 66: an average of 22.

Q.13 · Answer (C) 11 s

To cross completely the trains must close the gap and pass each other: 285 + 310 + 340 = 935 m. Relative speed = 151 + 155 = 306 km/h = 85 m/s, and 935 ÷ 85 = 11 s.

Q.14 · Answer (C) 11

(11 + x)/(29 + x) = 11/20 gives 220 + 20x = 319 + 11x, so 9x = 99 and x = 11.

Q.15 · Answer (B) 10

The volume is unchanged: $(1/3)\pi \times 3.5^2 \times 7.2 = 2 \times (1/3)\pi \times 2.1^2 \times h$, so 88.2 = 8.82h and h = 10 cm.

Q.16 · Answer (B) 15

4 kg in the ratio 4 : 1 : 3 is 2 kg of X, 0.5 kg of Y and 1.5 kg of Z: 2 × 2 + 0.5 × 4 + 1.5 × 6 = 4 + 2 + 9 = ₹15.

Q.17 · Answer (D) 350

SI = 2,500 × 7 × 2/100 = ₹350.

Q.18 · Answer (A) 21 kg

The total weight rises by 6 × 1.5 = 9 kg, so the new child weighs 12 + 9 = 21 kg.

Q.19 · Answer (C) 175.9

New total = 10 × 172 + 178 + 165 – 3 × 160 = 1,720 + 343 – 480 = 1,583 cm for 9 students: 1,583/9 ≈ 175.9 cm.

Q.20 · Answer (C) 700

SI = 2,500 × 7 × 4/100 = ₹700.

Reasoning Ability and Problem Solving

Q.1 · Answer (A) 20 + 5 × 4 = 85

Interchanging '+' and '×' and the numbers 4 and 5 turns 20 + 5 × 4 into 20 × 4 + 5 = 80 + 5 = 85. The other three give 65, 54 and 30.

Q.2 · Answer (A) 0

After interchanging the signs: 392 ÷ 7 + 2 × 12 – 80 = 56 + 24 – 80 = 0.

Q.3 · Answer (D) 9
Y is the only letter in both words and 9 the only digit in both codes, so Y = 9.

Q.5 · Answer (C) 222
The second number is the cube of the first plus 6: $4^3 + 6 = 70$ and $8^3 + 6 = 518$. Likewise $6^3 + 6 = 222$.

Q.6 · Answer (C) TGTFHEBXL
Each letter moves 7 places back, wrapping round from A to Z: Y → R, A → T, T → M, C → V, H → A. So ANAMOLIES → TGTFHEBXL.

Q.8 · Answer (A) 178
The differences are 4, 8, 12, 16, rising by 4, so the next is 20: 158 + 20 = 178.

Q.11 · Answer (A) WZ70
Each letter moves 3 places forward and the number is doubled: L → O, O → R, 15 → 30. So TW35 → WZ70.

Q.12 · Answer (D) Only conclusion (I) follows
As all arrows are swords, some swords are arrows: I follows. Nothing is said about dragons outside the swords, so II does not follow.

Q.13 · Answer (A) 1
Facing the centre, a person's right is anticlockwise and left clockwise. Clockwise the seats are E, A, C, B, D, F, G: G, F and D sit between E and B on E's right, C is on A's left, F sits between G and D, and D is not next to E. Seated alphabetically clockwise from A they would be A, B, C, D, E, F, G, so only D keeps his seat.

Q.14 · Answer (B) Only conclusion 1 follows
The nurses who are patients cannot be doctors, so some patients are not doctors: 1 follows. The patients who are medicines need not be nurses, so 2 does not follow.

Q.15 · Answer (C) G
Facing the centre, a person's left is clockwise and right anticlockwise. Clockwise the seats are F, B, G, E, D, C, A: G is second to F's left, F third to D's left, C third and A second to B's right, and E, in the one seat left, is not next to B. E's immediate right is G.

Q.16 · Answer (C) NLRP
Each letter moves 7 places forward: N → U, L → S, R → Y, P → W, and K → R, I → P, O → V, M → T. So GEKI → NLRP.

Q.17 · Answer (C) JVC
First letters move back 2 (R, P, N, L, J), second letters forward 3 (J, M, P, S, V) and third letters back 2 (K, I, G, E, C): JVC.

Q.18 · Answer (B) 5
N is the only letter in both words and 5 the only digit in both codes, so N = 5.

Q.19 · Answer (C) 6
S is the only letter in both words and 6 the only digit in both codes, so S = 6.

Q.20 · Answer (A) 7
The second number is twice the first plus 1: $2 \times 19 + 1 = 39$ and $2 \times 1 + 1 = 3$. Likewise $2 \times 3 + 1 = 7$.

General Awareness

Q.1 · Answer (B) Westerly jet stream
In winter the westerly jet stream, blowing south of the Himalayas, brings the western disturbances from the Mediterranean region to north and north-west India.

Q.2 · Answer (A) Bharatanatyam
SSC's key gives Bharatanatyam. Aparna Satheesan, who received the Abhinandan Saroja National Award for 2021, began with Bharatanatyam as a child and performs both it and Kuchipudi; profiles of her also name Mohiniyattam among her other forms.

Q.3 · Answer (C) Bihar and West Bengal
In the 2011 census Bihar, at 1,106 persons per km², was the most densely populated state, followed by West Bengal.

Q.4 · Answer (A) Jabalpur
The West Central Railway zone has its headquarters at Jabalpur, Madhya Pradesh.

Q.5 · Answer (B) Dantidurga
The Rashtrakuta founder Dantidurga performed the 'hiranya-garbha' (golden womb) ritual, which was thought to make the performer a Kshatriya by rebirth.

Q.6 · Answer (A) Speaker of Lok Sabha
Under Article 100 the Speaker (or whoever presides) does not vote in the first instance, but has a casting vote when the votes are equal.

Q.7 · Answer (B) Dreaming Big: My Journey to Connect India
Sam Pitroda's autobiography is 'Dreaming Big: My Journey to Connect India' (2015). The others are the autobiographies of Harivansh Rai Bachchan, Dilip Kumar and Kuldip Nayar.

Q.8 · Answer (B) 15th August
Under the Indian Independence Act, 1947, power passed on 15 August 1947, when the Constituent Assembly also became the legislature of the new Dominion.

Q.9 · Answer (A) Articles 25-28
Articles 25 to 28 make up the Right to Freedom of Religion.

Q.10 · Answer (C) Calcium Deficiency
Teeth are built largely of calcium phosphate, and milk is a chief source of calcium; a child who avoids milk may have too little calcium for the new teeth to form.

Q.11 · Answer (B) Mahatma Gandhi
Mahatma Gandhi launched the Quit India Movement in August 1942 with the call 'Do or Die'.

Q.12 · Answer (D) Ravneet Kaur
Ravneet Kaur, a 1988-batch IAS officer, took charge as Chairperson of the Competition Commission of India on 23 May 2023.

Q.13 · Answer (D) Vaisheshika - Vyasa
The Vaisheshika system is ascribed to Kanada; Vyasa (Badarayana) is linked with the Vedanta. The other three pairs are right.

Q.14 · Answer (D) Only II
Public goods are non-rivalrous and non-excludable, and serving one more user adds nothing to the cost (zero marginal cost). So only II is true.

Q.15 · Answer (B) Chhau
Chhau, the masked dance of Jharkhand, Odisha and West Bengal, was inscribed on UNESCO's Representative List in 2010.

Q.16 · Answer (B) New Delhi
The 2023 IBA Women's World Boxing Championships were held in New Delhi in March 2023.

Q.17 · Answer (A) 5.6
SSC's key gives 5.6 mm, the .22 calibre of the single-loaded small-bore rifles used in the 50 m rifle events. It is not strictly used in every event: the 10 m air rifle fires 4.5 mm pellets.

Q.18 · Answer (C) Iron
The Harappans did not know iron. They built with burnt bricks, used wood, and set the bricks of structures such as the Great Bath in gypsum mortar.

Q.19 · Answer (B) 2013
The National Food Security Act was enacted in 2013.

Q.20 · Answer (D) 137
Under the PM Formalisation of Micro food processing Enterprises (PMFME) scheme, the Ministry of Food Processing Industries has approved One District One Product for over 700 districts, covering 137 unique products.

Q.21 · Answer (A) Seismic Wave
The energy of an earthquake travels through and around the earth as seismic waves. A main shock is the largest quake of a sequence and a foreshock a smaller one before it.

Q.22 · Answer (A) Gandhi Irwin Pact
Under the Gandhi-Irwin Pact of March 1931 the Congress called off civil disobedience and agreed to attend the Second Round Table Conference, where Gandhi was its sole representative.

Q.23 · Answer (D) Sunlight
Plants capture sunlight in photosynthesis, and that energy passes along the food chains to the rest of the ecosystem.

Q.24 · Answer (A) Ustad Alladiya Khan
Ustad Alladiya Khan founded the Jaipur-Atrauli gharana of Hindustani vocal music.

Q.25 · Answer (B) 21.82%
Carbon in the CO_2 = (12/44) × 0.20 ≈ 0.0545 g, which is (12/44) × (0.20/0.25) × 100 ≈ 21.82% of the compound.

English Language and Comprehension

Q.1 · Answer (A) substantial
Donations that improved the community's quality of life were large: 'substantial'. 'Meagre' and 'parsimonious' mean scanty or stingy.

Q.2 · Answer (D) Faithfull
The correct spelling is 'faithful': the suffix '-ful' has one 'l'.

Q.3 · Answer (B) afraid
Fear of something is expressed with the adjective 'afraid of': 'She is afraid of spiders.'

Q.4 · Answer (A) To uncover negative information about someone or something.
To 'dig up dirt' on someone is to search out damaging information about them.

Q.5 · Answer (C) competed
A completed past event takes the simple past: 'the athlete competed'. 'Were' and 'have' do not agree with the singular 'the athlete'.

Q.6 · Answer (A) Obedence
The correct spelling is 'obedience'.

Q.7 · Answer (D) These are the works of the great poet Tagore.
SSC's key gives 'the great poet Tagore': a well-known person named with a descriptive title takes 'the'. 'An' is wrong before the consonant sound of 'great' and 'this' needs an earlier mention; 'that great poet' is an emphatic use also heard.

Q.8 · Answer (B) Boastful
Of the options only 'boastful' is opposite to 'modest', in its sense of unassuming. 'Humble', 'moderate' and 'unimpressive' are close to its senses, including the 'small' sense it has in 'a modest part'.

Q.9 · Answer (A) Independent
The correct spelling is 'independent'.

Q.10 · Answer (C) Just impossible
'When pigs fly' describes something that will never happen.

Q.11 · Answer (B) brake
A car is stopped with its 'brake'; 'break', which sounds the same, means to smash or a pause.

Q.12 · Answer (B) greedy
'Avaricious' means having an excessive desire for wealth: 'greedy'.

Q.13 · Answer (A) about the 1940s were quite
A period of time takes 'in': 'The residents of Rameswaram in the 1940s were quite orthodox'.

Q.14 · Answer (B) orchestra
A symphony is performed by an orchestra, and 'an' fits the vowel sound of 'orchestra'.

Q.15 · Answer (C) lie
'Lie' (to rest in a flat position) takes no object; 'lay' needs one, and 'lye' and 'lei' are nouns.

Q.16 · Answer (C) asleep
To 'fall asleep' is to begin sleeping; the adjective 'asleep' follows 'fell'.

Q.17 · Answer (B) a team successfully launched
The team is a particular one, the one that did the planning and whose product it was, so 'the team successfully launched'.

Q.18 · Answer (D) the exceptional movie
'Of all the movies' calls for a superlative: 'the most exceptional movie'.

Q.19 · Answer (D) Harmony
'Cacophony' is a harsh mixture of sounds; its opposite is 'harmony'. 'Discord', 'noise' and 'dissonance' are close to it in meaning.

Q.20 · Answer (B) The salesman insisted that the payment against the purchase should be done by using a bank card.
SSC's key gives the version with 'the salesman', 'the payment' and 'the purchase', all particular ones, and 'a bank card', any card. Two versions use 'an' before a consonant sound ('an purchase', 'an bank card'); the all-'a' version is grammatical too, but speaks of no particular salesman or purchase.

Q.21 · Answer (C) highest
Peaks are ranked by height: the world's ninth 'highest' peak.

Q.22 · Answer (B) ascent
An 'ascent' is a climb, here the route by which the summit was first climbed.

Q.23 · Answer (C) translates
A name 'translates as' its meaning in another language: Nanga Parbat means 'naked mountain'.

Q.24 · Answer (C) fatality rate
The share of climbers who die is the 'fatality rate'; 'mortal rate' and 'frailty rate' are not used for this.

Q.25 · Answer (C) exacerbated
'Exacerbated' means made worse: the climbers' misery is worsened by poor local support. The other options do not fit after 'is' and before 'by'.

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